

COMP20008 Elements of Data Processing



Lecture 1: Introduction to the Subject and Data Processing

Semester 2 2026

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What is Data Processing?

Why is it important?

Scientific Method

SCIENTIFIC METHOD

A JOURNEY OF DISCOVERY



- Data processing is a significant part of Science
- It is the practical mechanism that **turns observations into scientific evidence**
- In addition, in order to draw reliable conclusions, we need to adhere to principles of scientific method and make sure the conclusions are supported by solid experimental design and methodology!

SCIENTIFIC METHOD

A JOURNEY OF DISCOVERY



- **Hypothesis:** *more study time leads to higher exam scores*
- **Data Input:** collect student data
- **Data Cleanup:** fix inconsistencies; reduce noise
- **Data Interpretation:** calculate correlations or build a statistical model
- **Data Visualisation** visualise the relationship
- **Draw Conclusions:** decide whether the evidence supports the hypothesis



1960s

2020s

More specific example from my own research: what is Trauma

The patient suffered severe cranial trauma.

The cyclist was taken to hospital with head trauma.

The survivors received treatment for trauma after the earthquake.

The soldier experienced psychological trauma after combat.

She was traumatised after witnessing the attack.

The war created collective trauma across the entire community.

That exam was traumatic—I never want to do it again.

She described the betrayal at the end of the relationship as relational trauma.

My supervisor's negative feedback was traumatic



1960s

More specific example from my own research: what is Trauma

The patient suffered severe cranial trauma.
The cyclist was taken to hospital with head trauma.
The survivors received treatment for trauma after the earthquake.
The soldier experienced psychological trauma after combat.

She was traumatised after witnessing the attack.

Observation and H1: trauma has changed over time across the entire
from physical injury psychological/emotional, i.e.
became broader and milder

again.

She described the betrayal at the end
relationship as relational trauma.

My super

Prof Nick Haslam:
“Concept Creep”
Theory



2020s



1960s

More specific example from my own research: what is Trauma

Observation and H1: trauma has changed over time from physical injury psychological/emotional, i.e. became broader and milder

The survivors received treatment for trauma after the

Setup experiment & collect data: survey people or a collect text corpus with mentions of “trauma”

Cleanup and prepare (e.g., normalise) the data

She was traumatised after witnessing the attack

Create a **model** to evaluate the hypothesis: how to measure the concept “breadth” and severity?

That exam was traumatic—I never want to do it

Visualise the results and evaluate the statistical significance of the observations

relationship as relational trauma

Draw conclusions

2020s



1960s

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Draw conclusions

2020s

The real world is complex. We **reduce the complexity by making assumptions** and create a **model**!

Read the paper: Baes, N., Vylomova, E., Zyphur, M., & Haslam, N. (2023). [The semantic inflation of “trauma” in psychology](#). *Psychology of Language and Communication*, 27(1), 23-45.



1960s

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The survivors received treatment for trauma after the

Setup experiment & collect data: survey people or collect a text corpus with mentions of “trauma”

Cleanup and prepare (e.g., normalise) the data

Create a **model** to evaluate the hypothesis: how to measure the concept “breadth” and severity?

That exam was traumatic—I never want to do it

Visualise the results and evaluate the statistical significance of the observations

relationship as relational trauma

Draw conclusions

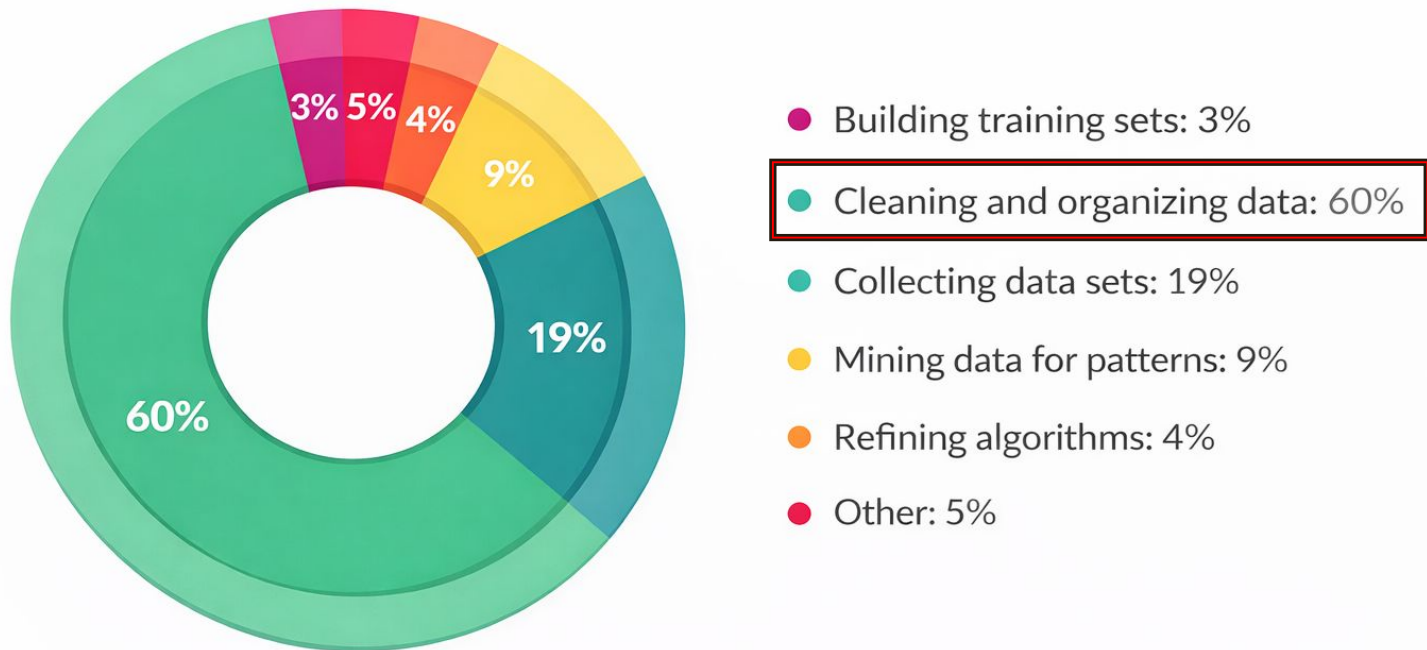
2020s

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Most time consuming parts

What data scientists spend the most time doing



Why is it **important**?

color	director_name	duration	gross	movie_title	language	country	budget	title_year	imdb_score
Color	Martin Scorsese	240	116866727	The Wolf of Wall Street	English	USA	100000000	2013	8.2
Color	Shane Black	195	408992272	Iron Man 3	English	USA	200000000	2013	7.2
color	Quentin Tarantino	187	54116191	The Hateful Eight	English	USA	44000000	2015	7.9
Color	Kenneth Lonergan	186	46495	Margaret	English	usa	14000000	2011	6.5
Color	Peter Jackson	186	258355354	The Hobbit: The Desolation of Smaug	English	USA	225000000	2013	7.9
	N/A	183	330249062	Batman v Superman: Dawn of Justice	English	USA	250000000	202	6.9
Color	Peter Jackson	-50	303001229	The Hobbit: An Unexpected Journey	English	USA	180000000	2012	7.9
Color	Edward Hall	180		Restless	English	UK		2012	7.2
Color	Joss Whedon	173	623279547	The Avengers	English	USA	220000000	2012	8.1
Color	Joss Whedon	173	623279547	The Avengers	English	USA	220000000	2012	8.1
	Tom Tykwer	172	27098580	Cloud Atlas	English	Germany	102000000	2012	-7.5
Color	Null	158	102515793	The Girl with the Dragon Tattoo	English	USA	90000000	2011	7.8
Color	Christopher Spencer	170	59696176	Son of God	English	USA	22000000	2014	5.6
Color	Peter Jackson	164	255108370	The Hobbit: The Battle of the Five Armies	English	New Zealand	250000000	2014	7.5
Color	Tom Hooper	158	148775460	Les Misérables	English	USA	61000000	2012	7.6
Color	Tom Hooper	158	148775460	Les Misérables	English	USA	61000000	2012	7.6

Why is it important?

- Either remove or fill in the missing values
- Correct misspelled words
- Fix data input errors
- Identify outliers

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Color	Martin Scorsese	240	116866727	The Wolf of Wall Street	English	USA	100000000	2013	8.2
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Data comes in different forms

en.wikipedia.org

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2022 West Java earthquake

From Wikipedia, the free encyclopedia

Coordinates: 6°53′S 107°09′E﻿ / ﻿6.883°S 107.095°E﻿ / -6.883; 107.095

On 21 November 2022, at 13:21 WIB (UTC+07:00), a M_{w} 5.6 earthquake struck near Cianjur in West Java, Indonesia. At least 310 people died, 2,046 were injured and 24 remain missing. More than 56,311 homes were damaged in Cianjur. The earthquake was strongly felt in Jakarta. It is the deadliest earthquake to hit Indonesia since the 2018 Sulawesi earthquake.^[1]

Contents [hide]

- Tectonic setting
- Earthquake
- Impact
- Casualties
 - Search and rescue
 - Aftermath
 - Reconstruction
- Response
- See also
- References
- External links

Tectonic setting

Java lies near an active convergent boundary Plate to the south. At the boundary, marked by subducts beneath the Sunda Plate. The subdu magnitude 8.7, while the Australian Plate may (intraslab earthquakes) beneath the coast of J and tsunamis in 2006 and 1994. An intraslab e

2022 West Java earthquake

Collapsed homes in Cianjur Regency.

Show map of Indonesia
Show map of Java
Show all

UTC time 2022-11-21 06:21:10
ISC event 625234413
USGS-ANSS ComCat
Local date 21 November 2022

	A	B	C	D
1	customer_id	order_value	date_ordered	delivery_state
2	55321	129.99	11/02/2018	VIC
3	55165	32.88		NSW
4		99.00	15/02/2018	VIC
5		88.98	17/02/2018	TAS
6	55193	\$52.00	18/02/2018	NSW
7	55113	15865	18/02/2001	SA
8	55155	132.95	19/02/2018	Victoria
9	55119	111.95	4.32E+04	QLD
10	-1	63.50	02/20/2018	N/A
11	55111	0.00	21/02/2018	VIC

PRID: [REDACTED] CMED CN: [REDACTED] Dispatch Number: [REDACTED]

Service: [REDACTED] EMS Date: May 6, 2013
Base: [REDACTED] EMS Team: ALS
Unit: 903 Crew 1: [REDACTED]
Shift: 6-11 SMT-P
EMD: Yes, With Pre-Arrival Instructions Crew 2: [REDACTED]
Dispatched As: Unconscious / Fainting * designates an ALS Provider
Mass Casualty: Not Applicable Mode to Rec: Lights / Sirens
Type of Scene: Unscheduled Moved From: Stretcher
Response Code: Alpha
Mode to Ref: No Lights/Sirens
Moved Via: Stretcher
Position: Supine
Outcome: Paramedic Response, Transport on same Service Ambulance
Ref Other Type: Doctor's Office Receiving: Hospital
Location: Surgical center Emergency Department
United States
Destination Basis: Patient's Physician's Choice

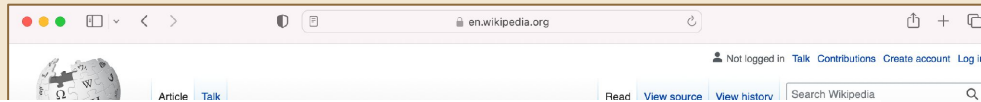
Last Name: [REDACTED] First: [REDACTED]
Address: [REDACTED]
City: [REDACTED] ST: [REDACTED] Zip: [REDACTED]
County: [REDACTED]
Country: United States
Citizenship: United States
DOB: [REDACTED] SSK: [REDACTED]
Age: Unknown Sex: M Weight: [REDACTED]

Odometer: [REDACTED] Times: [REDACTED]
Id Miles: 3.3 Dispatch: 08:44
EnRoute: 08:46
At Ref: 08:53
Leave Ref: 09:07
At Rec: 09:15
Available: 09:32
In Qtrs: 09:32
First Patch: 09:12
Medic onscene: 08:53

Notice of Privacy Practices Given: None
Consent Signed: No
PCS / Medical Necessity Signed: No

Scene Information
Scene: 1
Chief Complaint (Category: Unconscious / Fainting)
History of Present Illness
Surgical center pt is supine on operating table. Per staff pt went into a prgery was shocked twice and resumed sinus rhythm was amiodarone and started resusate with a strong radial pulse, b/p 240 /100 skin is warm and dry with intubated and being ventilated with a endtidal co2 of 35. noted good chest onlive bilateral iv access.
story Current Medications Allergies
None Listed None Listed
Neurological Exam
Glasgow Coma Scale
Unresponsive Loss of Consciousness: No
lyzed: Yes

Data comes in different forms



How to Read a 1940 U.S. Census Record PERSONAL INFORMATION

7	8	9	10	11	12	13	14	15	16
Shawburn, Thomas	Lodges	M	W	71	No	8	Illinois		
Name	Lodges	M	W	34	No	H3	Illinois		
Relationship to head of household	Lodges	M	W	25	No	CS	Illinois		
Wagner, John, A	Head	F	W	19	No	H4	Illinois		
Considine, Margaret	Head	F	W	55	No	8	Illinois		
Wagner, John, A	Sex	M	W	64	No	H2	Minnesota		
Wagner, Margaret	Sister	F	W	71	No	8	Minnesota		
Davis, George, A	Race	M	W	49	No	CS	Illinois		
Wagner, Margaret	Wife	F	W	44	No	CS	Illinois		
Wagner, Margaret	Age	4	S	4	No	8	Illinois		
Shaw, Bernard	Head	M	W	58	No	8	Illinois		
Lynman, Charles, H	Head	M	W	16	No	8	Illinois		
Anna, A	Wife	F	W	59	No	5	Illinois		
Lynman, William, D	Head	M	W	20	No	4	Illinois		
Hilda	Wife	F	W	48	No	8	Illinois		
Ruth, Amy	Daughter	F	W	22	No	8	Illinois		
James, W	Son	M	W	21	No	3	Illinois		
Leadbetter	Daughter	F	W						
Citizenship (blank if born in U.S.)									

FamilySearch.org

1940 U.S. CENSUS

earthquake



ordered delivery_state

02/2018 VIC

NSW

02/2018 VIC

02/2018 TAS

02/2018 NSW

02/2001 SA

02/2018 Victoria

.32E+04 QLD

02/2018 N/A

02/2018 VIC

PRID: [REDACTED]	CMD CN: [REDACTED]	Dispatch Number: [REDACTED]
Service: [REDACTED] EMS	Date: May 6, 2013	
Base: [REDACTED] EMS	Team: ALS	
Unit: 903	Crew 1: [REDACTED]	
Shift: 6-11	Crew 2: [REDACTED]	
EMD: Yes, With Pre-Arrival Instructions	Designates an ALS Provider	
Dispatched As: Unconscious / Fainting	Moved to Rec: Lights / Sirens	
Mass Casualty: Not Applicable	Moved From: Stretcher	
Type of Scene: Unscheduled		
Response Code: Alpha		
Moved to Ref: No Lights/Sirens		
Moved Via: Stretcher		
Position: Supine		
Outcome: Paramedic Response, Transport on same Service Ambulance		
Ref Other Type: Doctor's Office	Receiving: Hospital	
Location: Surgical center	Medical Center	
United States	Emergency Department	
	Destination Basis: Patient's Physician's Choice	
Last Name: [REDACTED]	First: [REDACTED]	Odometer
Address: [REDACTED]	City: [REDACTED] ST: [REDACTED] Zip: [REDACTED]	Id Miles: 3.3
Country: [REDACTED]	Country: United States	Dispatch: 08:44
Citizenship: United States	DOB: [REDACTED]	EnRoute: 08:46
Age: Unknown	Sex: M	At Ref: 08:53
	Weight: [REDACTED]	Leave Ref: 09:07
		At Rec: 09:15
		Available: 09:32
		In Qtr: 09:32
		First Patch: 09:12
		Medic onscene: 08:53
Notice of Privacy Practices Given: None		
Consent Signed: No		
PCS / Medical Necessity Signed: No		
Scene Information		
Scene: 1		
Chief Complaint (Category: Unconscious / Fainting)		
Arrest per staff		
History of Present Illness		
Surgical center pt is supine on operating table. Per staff pt went into a surgery was shocked twice and resumed sinus rhythm was amiodarone and started resuscitated with a strong radial pulse, b/p 240/100 skin is warm and dry with intubated and being ventilated with a tidal volume of 55. noted good chest auscultation bilaterally in access.		
History	Current Medications	Allergies
None Listed	None Listed	None Listed
Neurological Exam		
Response: Unresponsive	Loss of Consciousness: No	Glasgow Coma Scale
Analysed: Yes		

What is data?

Data is information that we record and use to turn into knowledge.

- Our aim: Wrangle/work with data
 - **Capture** and represent incoming data
 - **Turn it into information and knowledge**
 - **Compute** results with it and draw reliable conclusions

1. How to **get/capture** it?

2. How to **represent** it?

3. How to **transform** it?

4. How to **compute** with it?

Representing data in computers

Computers only work with binary data (0 s and 1 s)

Numeric

Numeric Values are stored in binary form.

Decimal 15 is mathematically

$$1 \times 10^1 + 5 \times 10^0$$

In **binary**, it is **1111**

$$1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

Textual

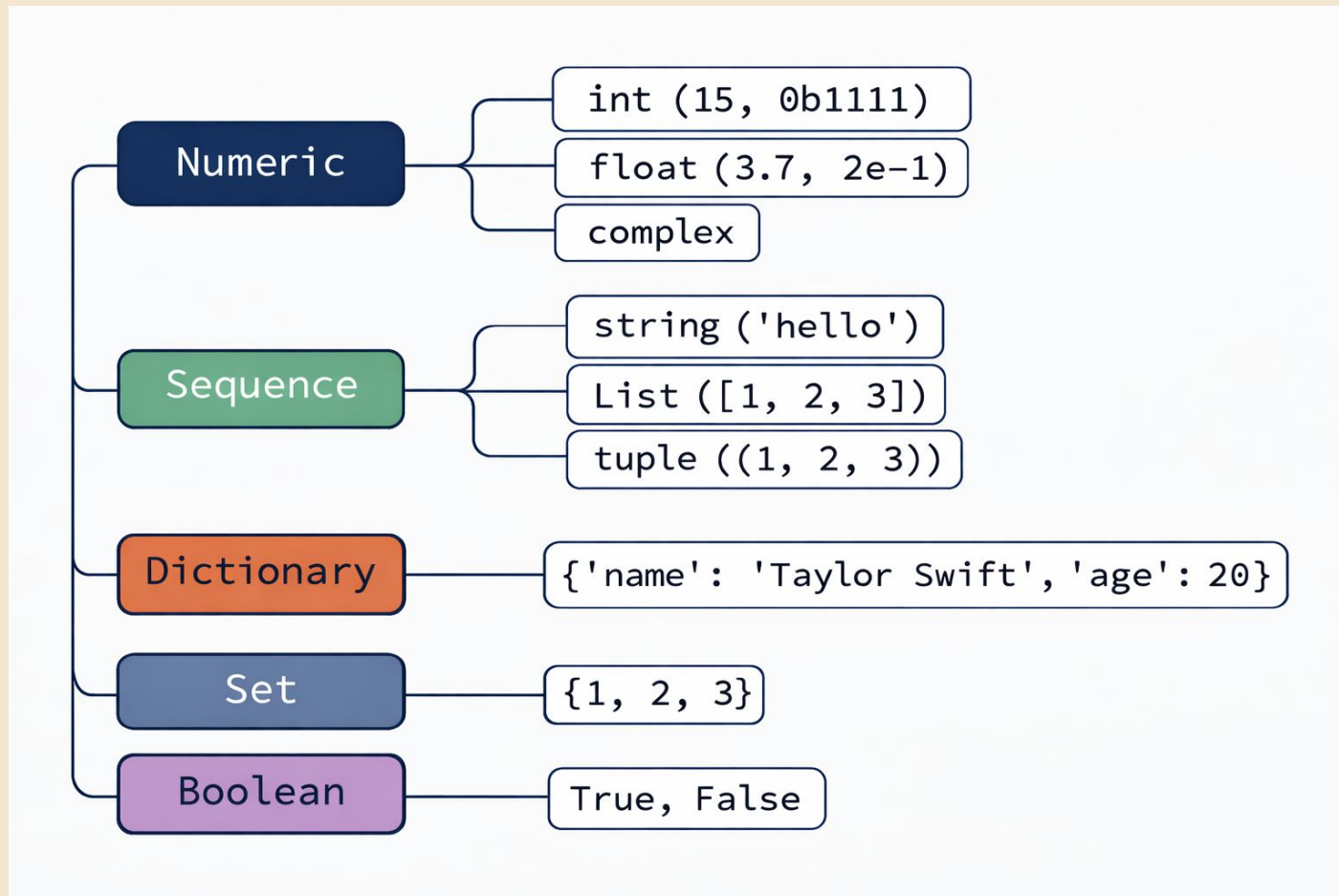
Textual data is also stored in binary form.

Each letter is a **different code point**:

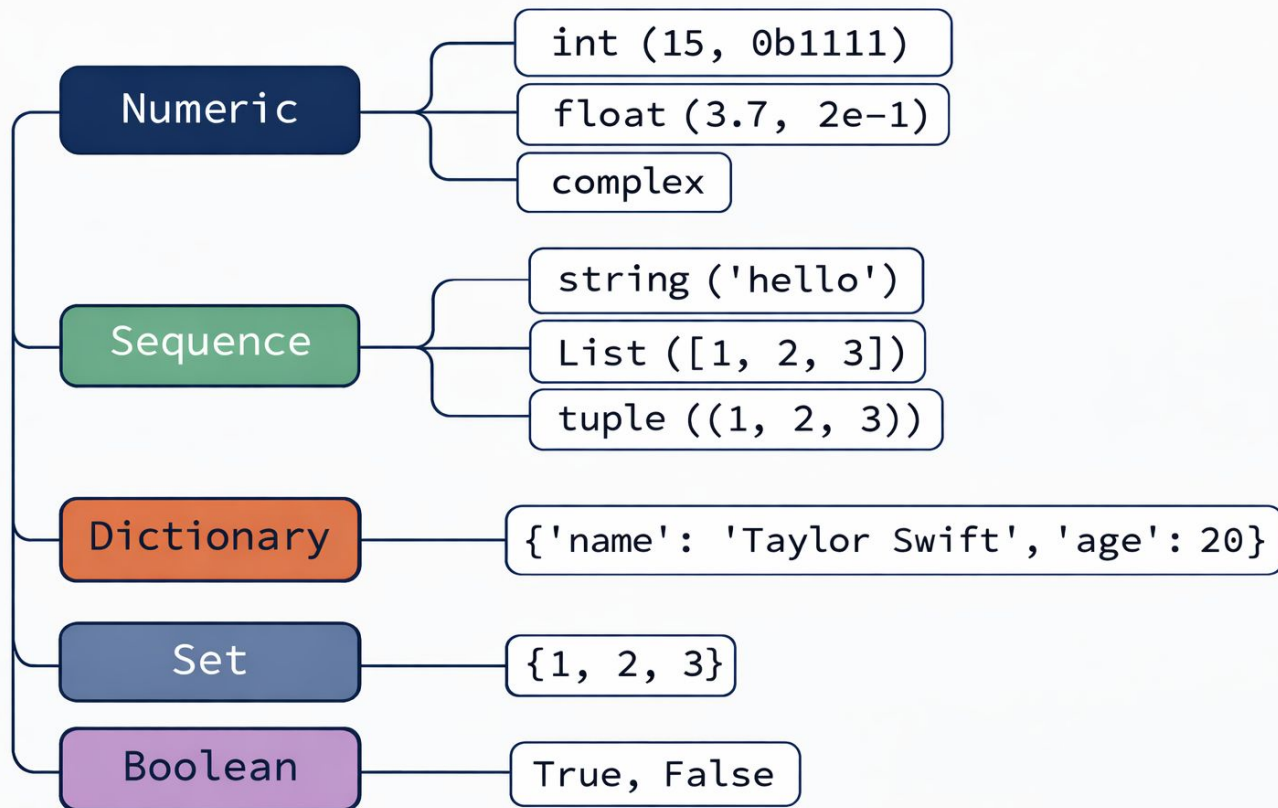
<https://www.unicode.org/charts/>

Letter	ASCII Code	Binary	Letter	ASCII Code	Binary
a	097	01100001	A	065	01000001
b	098	01100010	B	066	01000010
c	099	01100011	C	067	01000011
d	100	01100100	D	068	01000100
e	101	01100101	E	069	01000101
f	102	01100110	F	070	01000110
g	103	01100111	G	071	01000111

Basic data types in Python



Basic data types in Python



Recall: Immutable (read-only) vs mutable data types!

The Great Divide

Numerical Data (Quantitative)

- Integers or reals
- Can be **continuous**: Get the number from a function or from a sensor/reading of the world
- Can be **discrete**: Count the number
 - *Categorical: the numbers are just there: 1, 2*
 - *Ordinal: the numbers are ordered: 1st, 2nd*

Non-numerical data (Qualitative)

- Always discrete
- Made of Symbols (TRUE/FALSE, M/F) and Strings (“yes”, “John Smith”).

Dissecting Qualitative Data

Ordinal

(Ordered
Categorical)

Meaningful ordering, but no
meaningful distance measure (e.g.,
T-shirt sizes)



Nominal

Discrete names with no ordering
and no distance measure (e.g. City
names)

Includes a special limited type:
Categorical (e.g., True/False)



Dissecting Quantitative Data

There's no hard-and-fast rule. You can treat data both ways, it's a part of your model

Discrete

Countable items
(e.g. number of people in a city)
Restricted math operations (no averages)



Continuous

Uncountable, measurable items
(e.g. temperature, weight).
Uses full mathematical operations and distance measures.



The model design choice



T =

Discrete: 36, 37, 38, 39, ...

Continuous: 36.1, 37.5, 38.3, ...

Categorical: no_fever, mild_fever, severe_fever

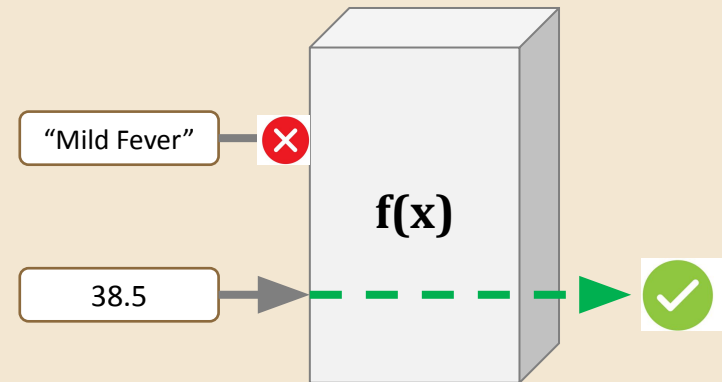
*Each choice has its pros and cons; and
it's your modelling part!*

The Transformation Problem

Warning!

Mathematical models **require numerical inputs**, but the real world provides a **massive amount of non-numerical (qualitative) data**.

How do we
bridge this gap?



Solution 1: Label Encoding

Maps category names to integer numbers

For example,

- Fever: 'mild'= 0, 'no'= 1, 'severe'= 2
- Diagnosis: 'cold'= 0, 'flu'= 1

Pros:

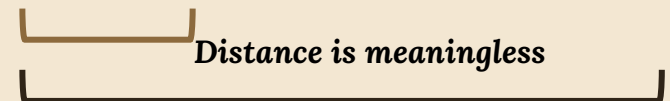
- Simplicity
- Space efficiency

Cons:

- Arbitrary ordinal relationships
- Distance is (often) meaningless

```
from sklearn.preprocessing import LabelEncoder  
label_encoder = LabelEncoder()  
encoded_data = label_encoder.fit_transform(data)
```

Red	Blue	Green	Orange	Pink
0	1	2	3	4

Distance is meaningless

Solution 2: One-hot encoding

```
from sklearn.preprocessing import OneHotEncoder
onehot_encoder = OneHotEncoder()
encoded_data = onehot_encoder.fit_transform(data)
```

Convert each category to a binary vector, where each element signals the presence or absence of that category

Pros

Avoids introducing an arbitrary ordinal relationships and arbitrary distance.

Cons

Greatly increases dimensionality resulting in sparse data (lots of zeros).

	Red	Blue	Green	Orange	Pink
Green	0	0	1	0	0
Blue	0	1	0	0	0

Red	Blue	Green	Orange	Pink
[1,0,0,0,0]	[0,1,0,0,0]	[0,0,1,0,0]	[0,0,0,1,0]	[0,0,0,0,1]

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Red	Blue	Green	Orange	Pink
[1,0,0,0,0]	[0,1,0,0,0]	[0,0,1,0,0]	[0,0,0,1,0]	[0,0,0,0,1]

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	Red	Blue	Green	Orange	Pink	Black
Green	0	0	1	0	0	0
Black	0	0	0	0	0	1
Grey	0	0	0	0	0	0

??

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	Red	Blue	Green	Orange	Pink	Black	Grey
Green	0	0	1	0	0	0	0
Black	0	0	0	0	0	1	0
Grey	0	0	0	0	0	0	1

... (and many many more...)

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Pros

Avoids introducing an arbitrary ordinal relationships and arbitrary distance.

Cons

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Good to consider the nature of the task: e.g., for colors the RGB model is more representative of the object nature

	Red	Blue	Green	Orange	Pink	Black	Grey
→	0	0	1	0	0	0	0
→	0	0	0	0	0	1	0
→	0	0	0	0	0	0	1

... (and many many more...)

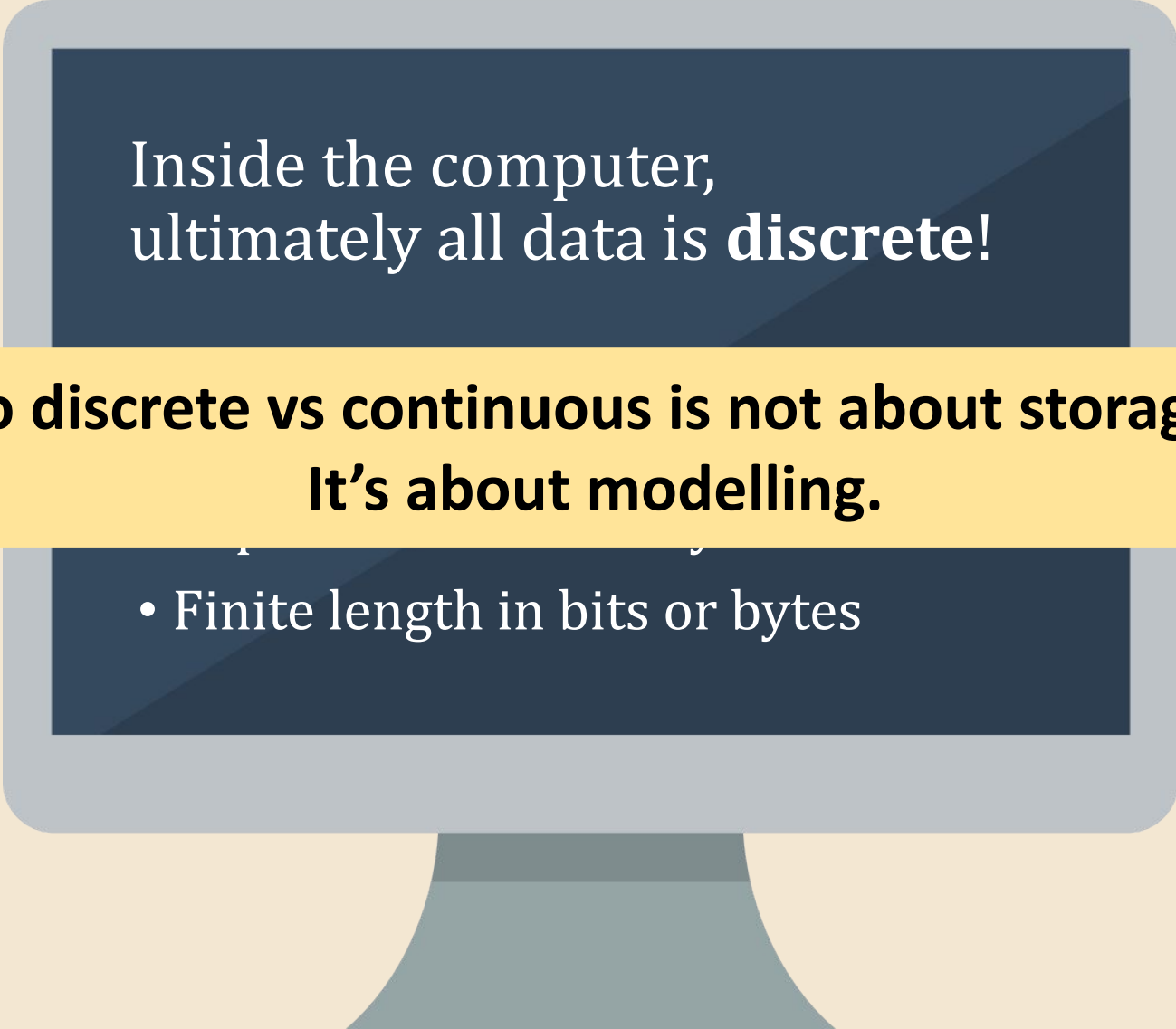
Discrete vs continuous: summary

Inside the computer,
ultimately all data is **discrete**!

Why?

- Represented in binary
- Finite length in bits or bytes

Discrete vs continuous: summary



Inside the computer,
ultimately all data is **discrete!**

**So discrete vs continuous is not about storage.
It's about modelling.**

- Finite length in bits or bytes

Discrete or continuous?

There is no HARD boundary.

Discrete vs Continuous is:

- **Not** about the object
- **Not** about binary storage
- **About** how you **model the variable**

Final Check: What about **Age**? Discrete or Continuous?

Discrete or continuous?

There is no HARD boundary.

Discrete vs Continuous is:

- **Not** about the object
- **Not** about binary storage
- **About** how you **model the variable**

Final Check: What about **Age**? Discrete or Continuous?

In principle:

You are getting older every millisecond → **continuous**

In practice:

“I am 20 years old.” → **discrete**

Representing Numerical Data

Number systems

- binary, decimal, octal, hexadecimal
- symbols:
 - Binary (base 2): 0, 1
 - Decimal (base 10): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
 - Octal (base 8): 0, 1, 2, 3, 4, 5, 6, 7
 - Hexadecimal (base 16): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- The value 15 is
 - $1111_2 = 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
 - $17_8 = 1 \times 8^1 + 7 \times 8^0$
 - $F_{16} = 15 \times 16^0$
- $4EBA_{16} = 4 \times 16^3 + 14 \times 16^2 + 11 \times 16^1 + 10 \times 16^0 = 20154$

(E)(B)(A)

Representing Non-numerical Data

For text, assign each character a number

ASCII code (outdated)

- 128 characters for English
- e.g., 011000001 is the number assigned to 'a'

Unicode

- UTF16, UTF8
 - Larger code space for different languages.
 - e.g., U+006F is the character 'o'
 - For the character '人', the Unicode is U+4EBA

Sets and sequences of data items

- Lists or sequences (vectors) of individual items
- Items of all kinds, sometimes even mixed
 - Text: list of words
 - Time series: list of (timepoint, value) pairs
 - DNA sequence: list of symbols A,T,G,C

Week 1 Workshop & Lecture 1 Alignment

Week 1 Workshop Goal

- Load structured data (CSV) using Pandas
- Manipulate tabular data (filter, sort, group, aggregate)
- Compute summary statistics (e.g., occupancy rate)
- Understand Series and DataFrame structures

Related Lecture 1 Slides (Part 2)

- The Core of Data Processing
Workshop practices *Data Input* and *Data Cleanup & Manipulation* in the pipeline.
- Data Types (Numerical vs Non-numerical)
Workshop applies these concepts to real columns (e.g., `tickets_sold` = numerical, `classification` = categorical).
- Discrete vs Continuous (Modelling)
Workshop models counts and rates (e.g., occupancy rate) highlighting that discrete vs continuous is about modelling.